

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43
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REG. NO. : 192372098

COURSE OUTCOME COVERED

CO -2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Design Task

Weightage: 20%

QUESTIONS:

Total Marks: 20

Question 1: Responsive Navigation Bar Design - Marks(4)

Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.

- (a) Identify key components required in a navigation bar for an e-commerce website.
 - (b) Design a responsive navigation bar using appropriate CSS techniques.
 - (c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).
 - (d) Evaluate the usability and suggest improvements for better user experience.
-

Question 2: Product Hover Effect - Marks(4)

Suppose that an e-commerce website wants to enhance user interaction through visual effects on product items.

- (a) Identify suitable CSS properties for hover effects.
- (b) Suggest appropriate **CSS animation or transition** for product interaction.
- (c) Design a hover effect for a product card.
- (d) Evaluate performance and user experience considerations.

Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.

- (a) Identify layout requirements of a dashboard (rows, columns, alignment).
 - (b) Compare **Flexbox and Grid** based on layout needs.
 - (c) Choose the most suitable layout technique and justify your choice.
 - (d) Evaluate the scalability and responsiveness of the chosen approach.
-

Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.

- (a) Identify key sections of a blog layout.
 - (b) Design a mobile-first layout using CSS.
 - (c) Demonstrate how the layout scales for larger screens using media queries.
 - (d) Evaluate accessibility and readability of the design.
-

Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

Suppose that a website includes a dynamic menu that responds to user interactions.

- (a) Identify appropriate JavaScript events for menu interaction.
- (b) Design event handling logic for menu toggle functionality.
- (c) Implement interaction for user actions (click/hover).
- (d) Evaluate performance and suggest improvements.

Code of Conduct:

I E.HEMANTH REDDY /192372098(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: E.HEMANTH Reddy

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
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Understanding of Concept	25	Demonstrates complete understanding of design requirements and concepts	Good understanding with minor gaps	Basic understanding with some errors	Poor understanding or incorrect concepts
Design / Implementation	30	Well-structured, efficient, and responsive design implementation	Correct implementation with minor issues	Basic implementation with inconsistencies	Incorrect or incomplete implementation
Use of Tools (CSS/JS)	20	Effective and advanced use of tools/techniques	Appropriate use of tools	Limited or inconsistent use	Incorrect or minimal use
Analysis & Evaluation	15	Insightful analysis with strong justification and optimization	Good analysis with justification	Basic analysis with limited depth	Poor or no analysis
Documentation / Clarity	10	Clear, well-structured, and properly presented solution	Mostly clear with minor issues	Basic clarity, missing details	Poor or unclear documentation

ANSWERS

Question 1

Responsive Navigation Bar Design

Key Components of an E-commerce Navigation Bar

A responsive e-commerce navigation bar should contain:

- Company Logo
- Home
- Categories/Products
- Search Bar
- Cart Icon
- Wishlist
- User Login/Profile
- Contact Us
- Hamburger Menu (for mobile devices)

Navigation Bar Layout

Desktop View

```
+-----+
| LOGO | Home | Products | Categories | Search | Cart | Wishlist | Login |
+-----+
```

Tablet View

```
+-----+
| LOGO | Home | Products | Search | Cart | Login |
+-----+
```

Mobile View

```
+-----+
| LOGO      ≡ Menu |
+-----+
```

When ≡ is clicked

+-----+		
Home		
Products		
Categories		
Search		
Cart		
Wishlist		
Login		
Contact		
+-----+		

CSS Techniques Used

- Flexbox
- Media Queries
- Responsive Width
- Box Model
- Hover Effects
- Mobile-first approach

Responsive Navigation (Hamburger Menu)

Desktop

LOGO Home Products Categories Cart Login

Mobile

LOGO ≡

After Clicking ≡

Home
Products
Categories
Cart
Wishlist
Login
Contact

(d) Evaluation & Improvements

Advantages

- Easy navigation
- Mobile friendly

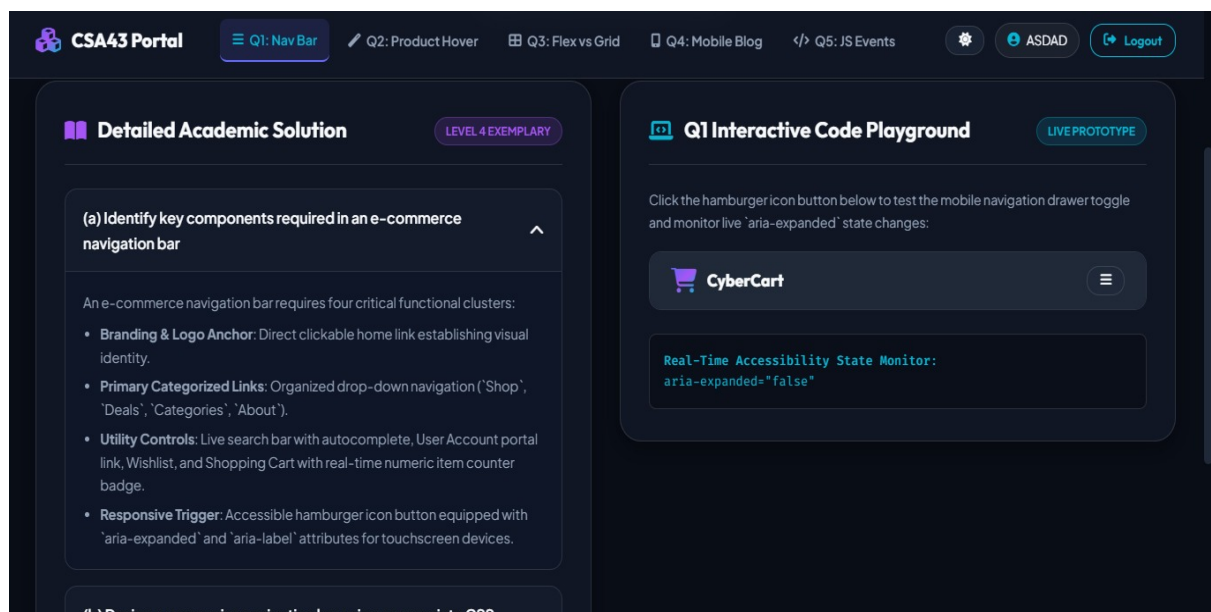
- Fast access to products
- Better user experience
- Supports different screen sizes

Improvements

- Add sticky navigation bar.
- Include search suggestions.
- Use icons with menu items.
- Add smooth animations for the hamburger menu.
- Highlight the active page.
- Ensure buttons are large enough for touch interaction.
- Optimize loading speed by minimizing CSS and JavaScript.

Conclusion

A responsive navigation bar built with **HTML, CSS Flexbox, and Media Queries** ensures that the e-commerce website works efficiently on desktop, tablet, and mobile devices. The hamburger menu improves usability on small screens, while responsive design techniques provide a consistent and user-friendly experience across all devices.



Question 2

Product Hover Effect

Suitable CSS Properties for Hover Effects

The following CSS properties create attractive hover effects:

- transform
- transition
- box-shadow
- opacity
- background-color
- color
- cursor

CSS Animation / Transition

```
.product{
  width:250px;
  border:1px solid #ccc;
  padding:15px;
  text-align:center;
  transition:0.4s ease;
}
.product:hover{
  transform:scale(1.05);
  box-shadow:0 8px 20px rgba(0,0,0,0.3);
}
```

Product Card Design

HTML

```
<div class="product">
  
  <h3>Sports Shoes</h3>
  <p>$50</p>
  <button>Buy Now</button>
</div>
```

Layout

```
+-----+
|   Product Image   |
|                   |
|   Sports Shoes    |
|   $50             |
|   [ Buy Now ]     |
```

+-----+

Hover

- ✓ Card enlarges
- ✓ Shadow appears
- ✓ Smooth animation

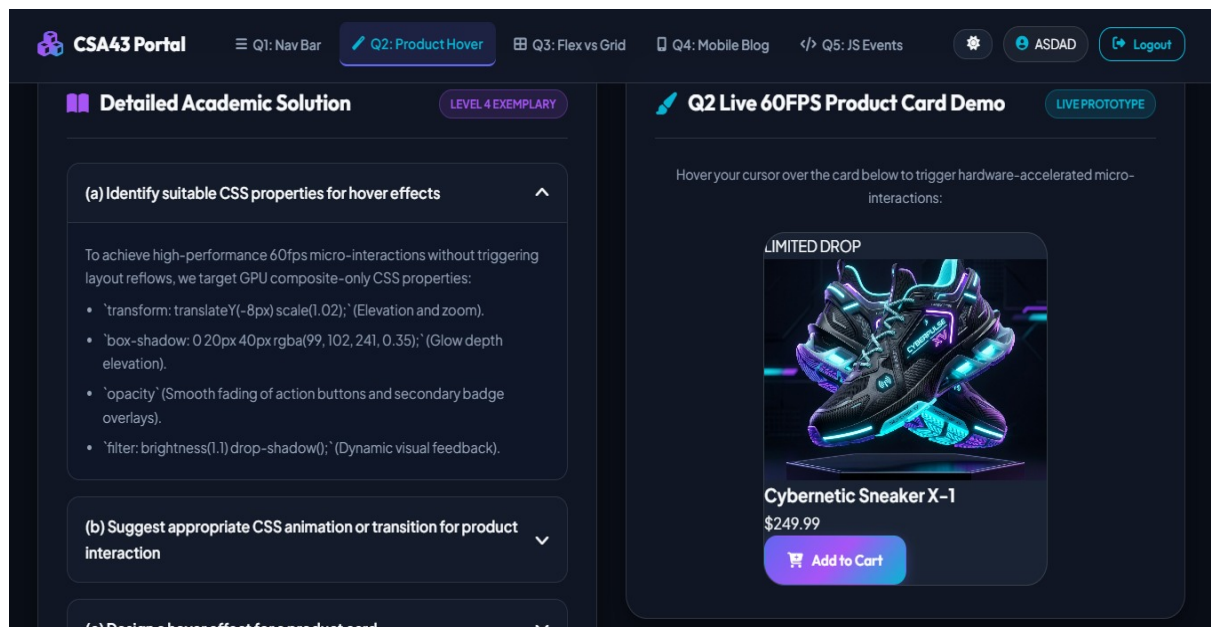
Evaluation

Advantages

- Improves product visibility.
- Attracts user attention.
- Gives interactive feedback.
- Enhances shopping experience.

Improvements

- Use smooth transitions.
- Avoid excessive animations.
- Optimize images.
- Keep hover effects lightweight.
- Ensure accessibility for keyboard users.



Question 3

Layout Selection (Flexbox vs Grid)

Dashboard Layout Requirements

A dashboard requires:

- Header
- Sidebar
- Main Content
- Widgets
- Footer
- Proper row and column alignment

Dashboard Layout

```
+-----+
|      HEADER      |
+-----+
| Sidebar | Dashboard | Statistics | Reports |
|         | Widget 1  | Widget 2  | Widget 3 |
+-----+
|      FOOTER      |
+-----+
```

Flexbox vs Grid

Feature	Flexbox	Grid
Layout	One-dimensional	Two-dimensional
Direction	Row or Column	Rows and Columns
Best For	Navigation bars	Complete dashboards
Alignment	Easy	Advanced
Complexity	Simple	Complex

Most Suitable Technique

CSS Grid is the best choice because:

- Controls both rows and columns.
- Easy to create dashboard layouts.
- Supports responsive design.
- Organizes widgets efficiently.

Example:

```
.dashboard{
display:grid;
```

```
grid-template-columns:250px 1fr 1fr;
gap:20px;
}
```

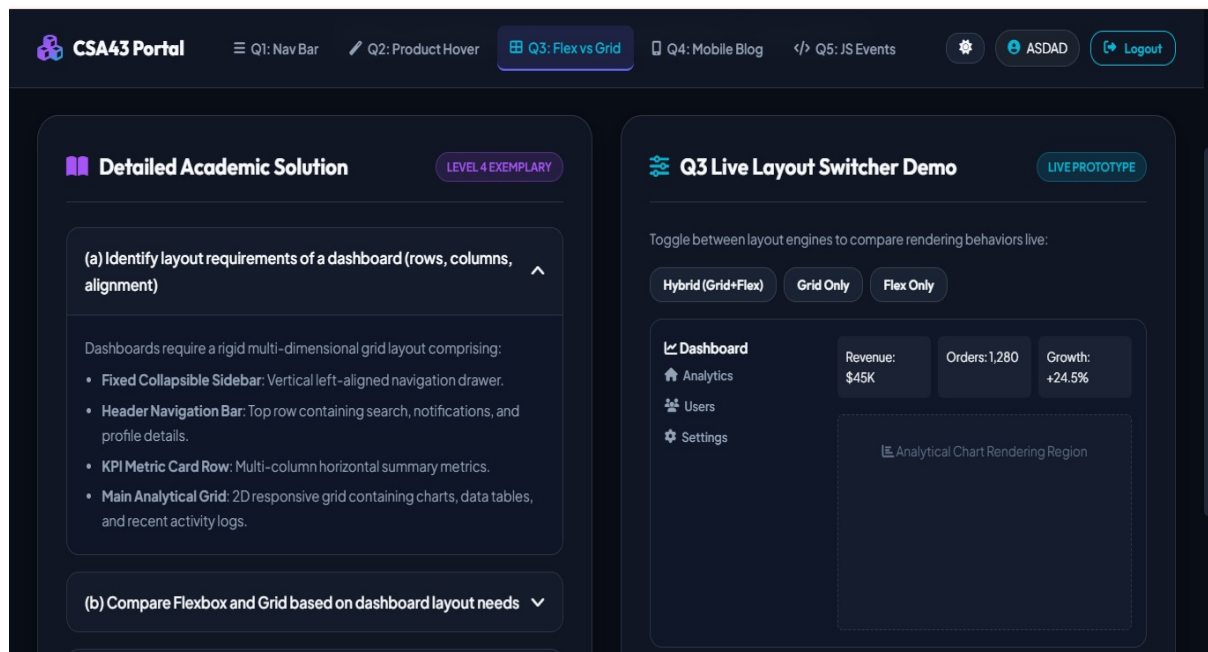
Evaluation

Advantages

- Highly scalable.
- Responsive on all devices.
- Easy maintenance.
- Efficient space utilization.

Improvements

- Combine Grid with Flexbox.
- Add media queries.
- Optimize widget sizes.
- Use auto-fit and minmax().



Question 4

Mobile-First Blog Layout

Key Sections

- Header
- Navigation
- Featured Image
- Blog Content
- Sidebar

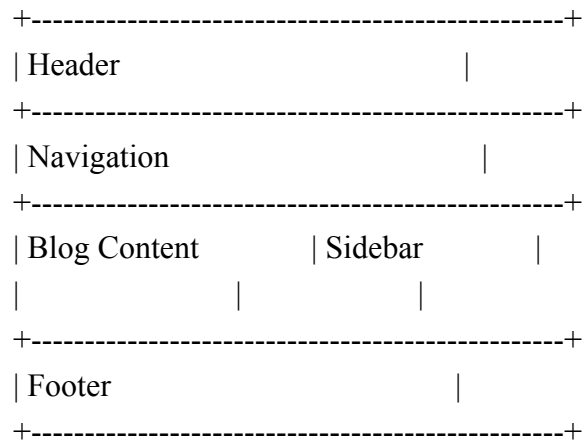
- Comments
- Footer

Layout

Mobile



Desktop



CSS Design

```

.container{
display:flex;
flex-direction:column;
}
@media(min-width:768px){
.container{
flex-direction:row;
}
}
  
```

```
}  
}
```

Media Query

```
@media(min-width:992px){  
.sidebar{  
width:30%;  
}  
.content{  
width:70%;  
}  
}
```

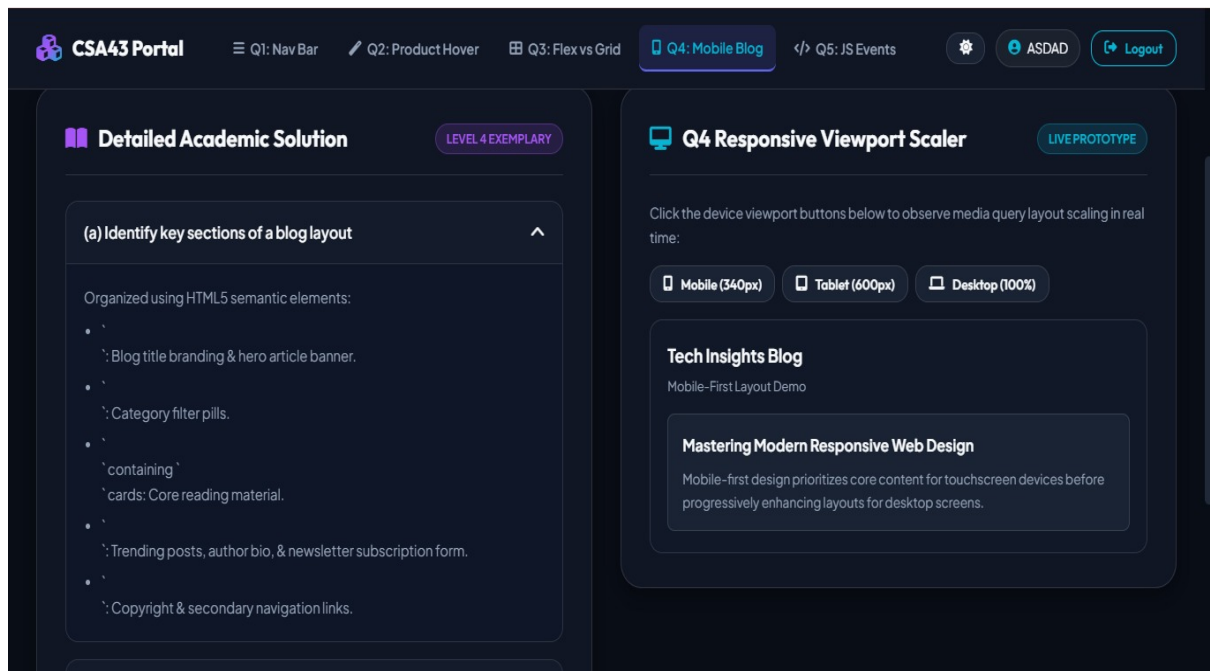
Evaluation

Advantages

- Mobile-friendly.
- Better readability.
- Faster loading.
- Responsive design.

Improvements

- Use larger fonts.
- Increase spacing.
- Optimize images.
- Add dark mode.
- Ensure keyboard accessibility.



Question 5

JavaScript Event Handling for Dynamic Menu

JavaScript Events

Common events:

- onclick
- onmouseover
- onmouseout
- onmouseenter
- onmouseleave

Event Handling Logic

Flow:

User Clicks ≡



JavaScript Function Executes



Menu Opens / Closes





Navigation Display Updated

Menu Toggle Implementation

HTML

```
<button onclick="toggleMenu()">≡</button>
<ul id="menu">
<li>Home</li>
<li>Products</li>
<li>About</li>
<li>Contact</li>
</ul>
```

CSS

```
#menu{
display:none;
}
```

JavaScript

```
function toggleMenu(){
let menu=document.getElementById("menu");
if(menu.style.display=="block")
menu.style.display="none";
else
menu.style.display="block";
}
```

Evaluation

Advantages

- Fast interaction.
- Easy navigation.
- Suitable for mobile devices.
- Reduces screen clutter.

Improvements

- Use CSS transitions for smoother animation.
- Debounce rapid clicks.
- Add keyboard navigation support.

- Optimize JavaScript performance.
- Improve accessibility with ARIA attributes.

 **CSA43 Portal**

Q1: Nav Bar

Q2: Product Hover

Q3: Flex vs Grid

Q4: Mobile Blog

Q5: JS Events

ASDAD

Logout

Detailed Academic Solution

LEVEL 4 EXEMPLARY

(a) Identify appropriate JavaScript events for menu interaction

Five key client-side JavaScript event listeners enable comprehensive accessibility:

- `click`: Primary mouse & touch state toggle.
- `keydown`: Accessible keyboard navigation ('Escape' key close, 'Enter' / 'Space' toggle).
- `touchstart` / `pointerdown`: Touch device rapid response eliminating 300ms click delay.
- `focusin` / `focusout`: Managing dropdown focus states.

(b) Design event handling logic for menu toggle functionality

Q5 Real-Time JS Event Console Logger

LIVE PROTOTYPE

Event Listener Activity Console

Clear Log

[System Ready] Click buttons or press keys to observe real-time JavaScript events ...

[14:56:04] [KEYDOWN]: Pressed Key: "Meta" (Code: MetaLeft)

[14:56:05] [KEYDOWN]: Pressed Key: "Shift" (Code: ShiftLeft)

Trigger Click Event

Toggle Menu State